

CSuprem Tutorial

(2D)



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Contents

- **CSuprem / Apsys introduction**
- **Getting started**
- **CSuprem commands**
- **A simple nMOS process**
- **Device simulation**

What is Suprem4?

- Stanford's 2D code 1970s to 1990s (up to 1 um technology).
- Intension: to cover process steps from substrate all the way to full device structure.
- Deposit, etch, implant, diffusion, oxidation, stress analysis.
- Core model: dopant diffusion (point defect model).
- Intended to interface to Pisces-2B (Stanford device 2D simulator).

What is Csuprem?

- Extension of technology to below 1 um (low energy implant models) while maintaining full compatibility.
 - Extension of Stanford's code to full 3D with inter-plane coupling.
 - Direct use of existing 2D input decks in 3D simulation.
 - Full 3D model for implantation, diffusion, segregation & oxidation.
 - Direct conversion of GDSII file into 3D simulation input decks.
 - Full 3D simulation for mechanical stress, bending and vibration motions.
 - Flexibility of switching between quasi-3D and full-3D modes according to speed/accuracy requirements.
- Extension of Stanford's code to full 3D with inter-plane coupling.



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Getting started

- Launch program.
- Various modules and subprograms:
SimuCSuprem, main program, CrosslightView.
- Locations of various control files.
- Running a simple example.
- Use of EasySetup2D.



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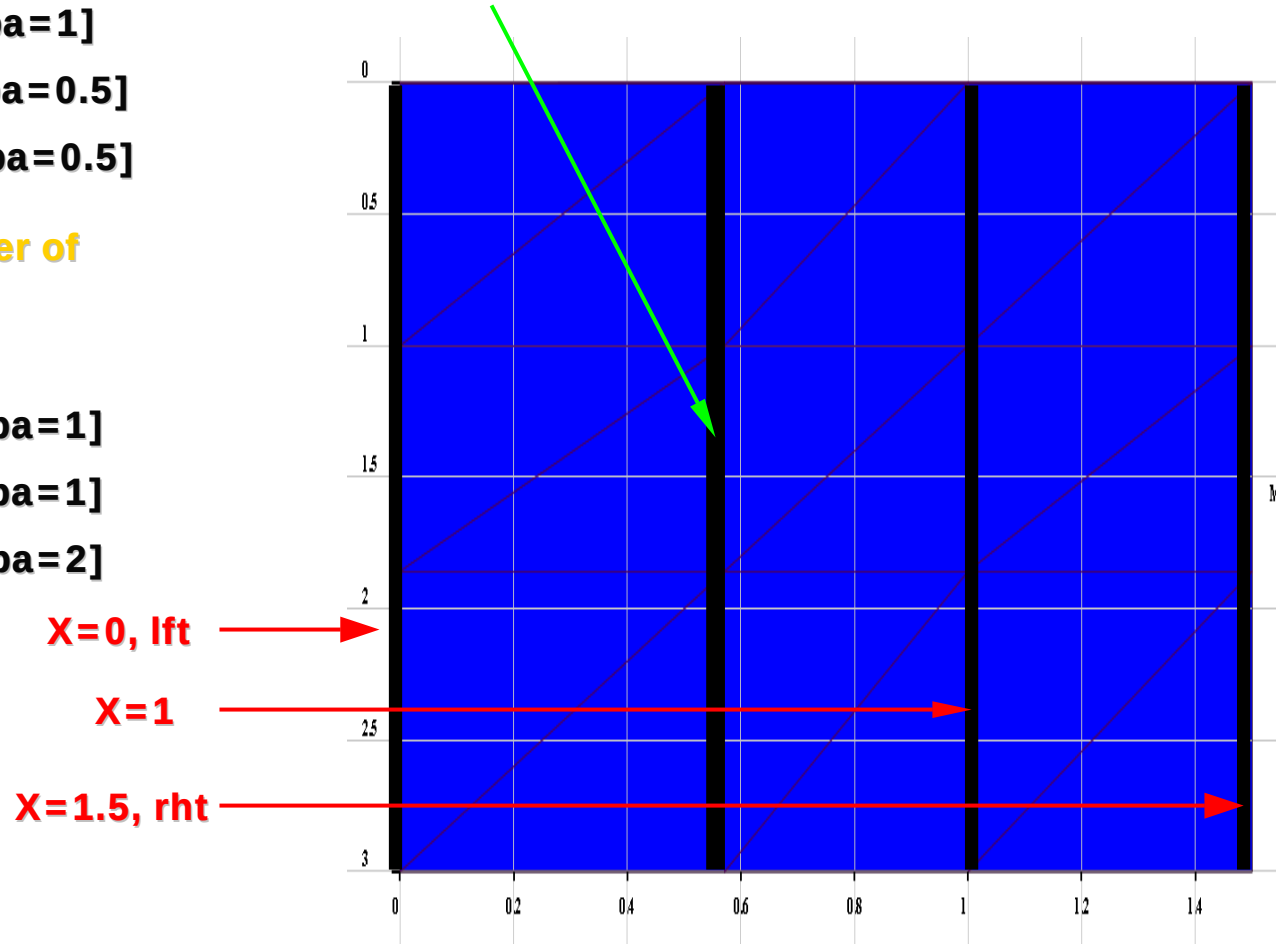
Line

line x loc=0.0 tag=left [spa=1]
 line x loc=1.0 [spa=0.5]
 line x loc=1.5 tag=right [spa=0.5]

**Default spacing: shorter of
two line distances**

line y loc=0.0 tag=top [spa=1]
 line y loc=1.0 [spa=1]
 line y loc=3.0 tag=bottom [spa=2]

Added line according to spacing

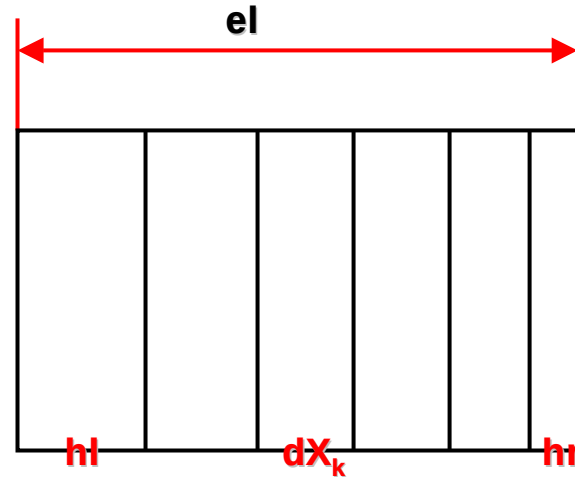


Line: Spacing

$$r = dx_{k+1} / dx_k < 1.5$$

$$el = dx_1 + dx_2 + \dots + dx_n$$

$$dx_n = r^{**n} * hl$$



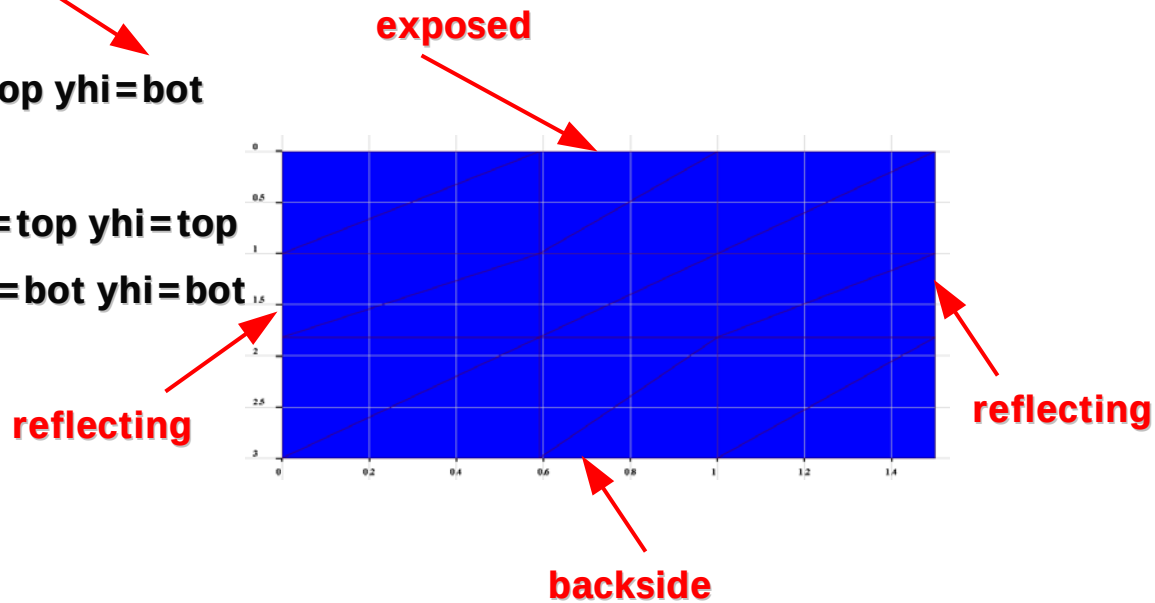
Region, Boundary

Tag defined by Line

material
 region silicon xlo=lft xhi=rht ylo=top yhi=bot

bound exposed xlo=lft xhi=rht ylo=top yhi=top
bound backside xlo=lft xhi=rht ylo=bot yhi=bot

Note: Y axis is flipped



Exposed: deposition, etch, oxidation, defect recombination and generation

Backside: defect recombination and generation

Reflecting: lateral implant scattering mirrors



Initialize

init boron conc=1.0e16 [orient = 100]

orient: substrate crystal orientation, default 100

boron : background doping type

conc: 1/cm³ background concentration

**Initialize sets up grid and background doping level.
(impurity is activated)**

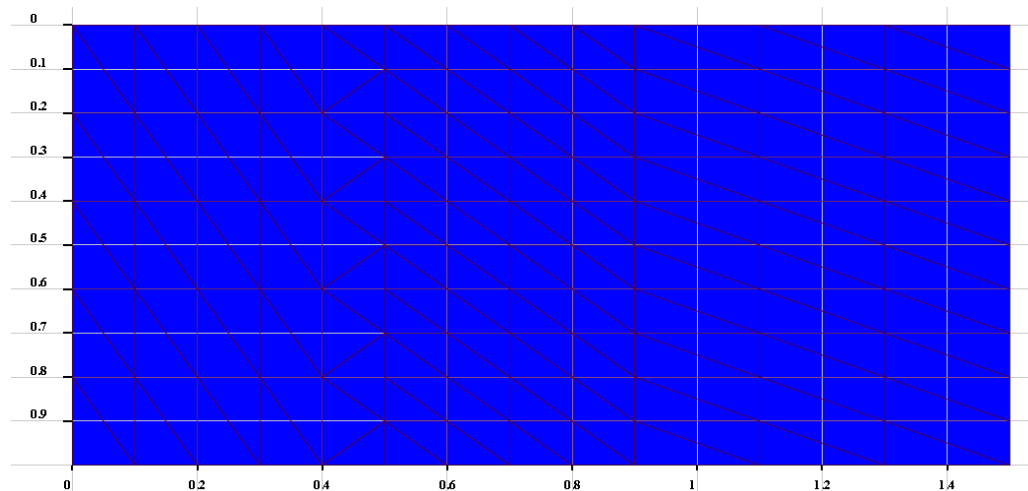
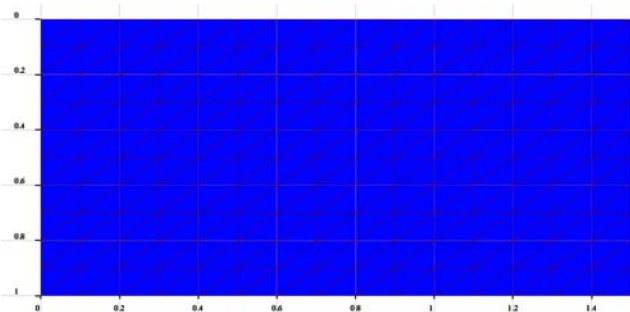
Eliminate – Reducing Mesh

elimin x.dir xlo=0 xhi=0.5 ylo=0 yhi=1 ntimes=1

elimin y.dir xlo=1 xhi=1.5 ylo=0 yhi=1 ntimes=1

Remove every other line

original

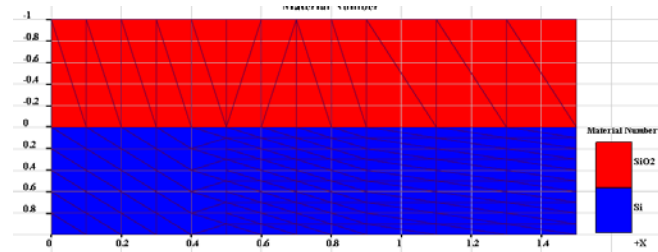


Elim by x.dir

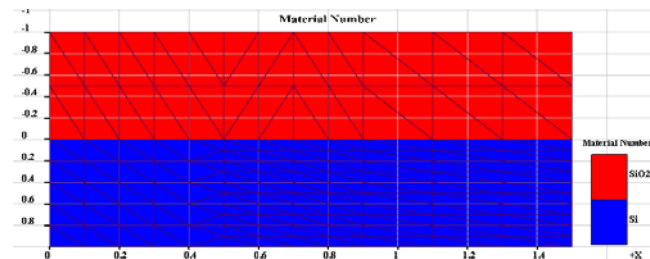
Elim by y.dir

Deposit

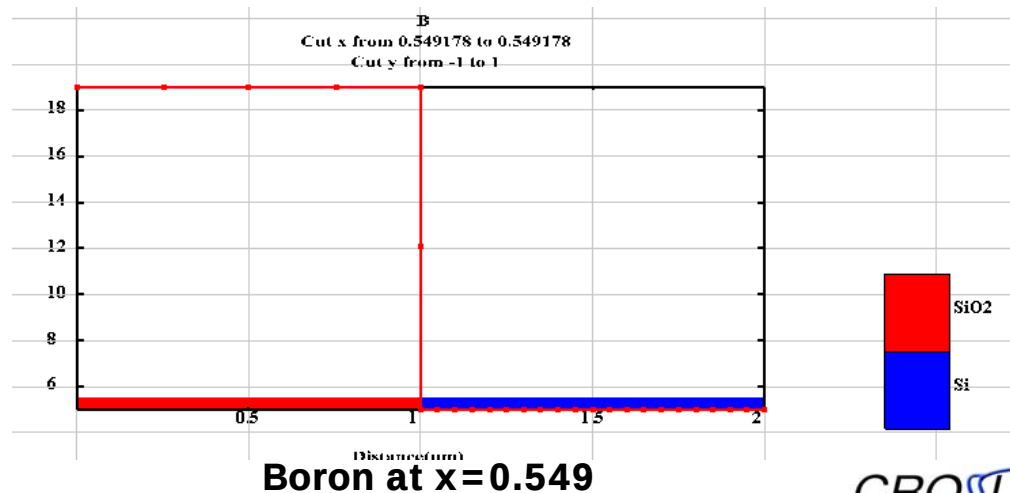
deposit oxide thick=1 [meshlayer=1]



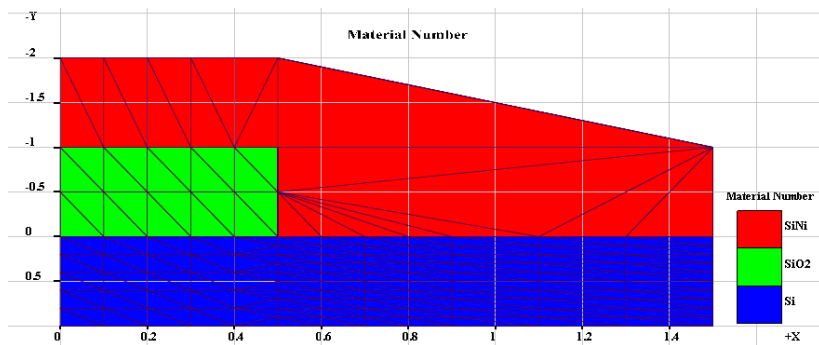
deposit oxide thick=1 meshlayer=2



deposit oxide thick=1 boron
conc=1.e+19 meshlayer=2

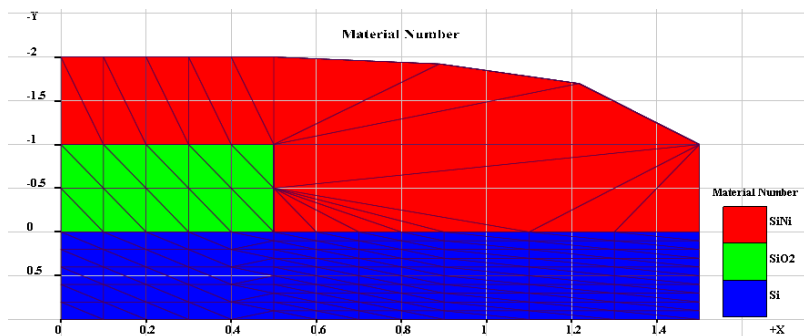


Deposit: Space

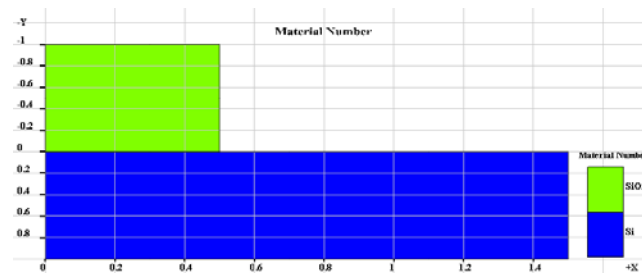


deposit nitride thick=1 **[space=1]**

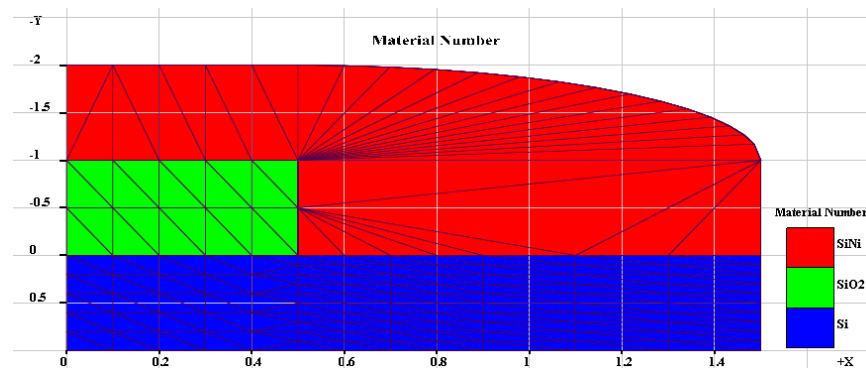
thick / meshlayer



deposit nitride thick=1 **space=0.4**



original

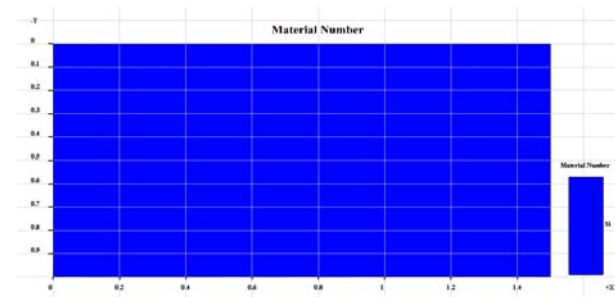


deposit nitride thick=1 **space=0.1**

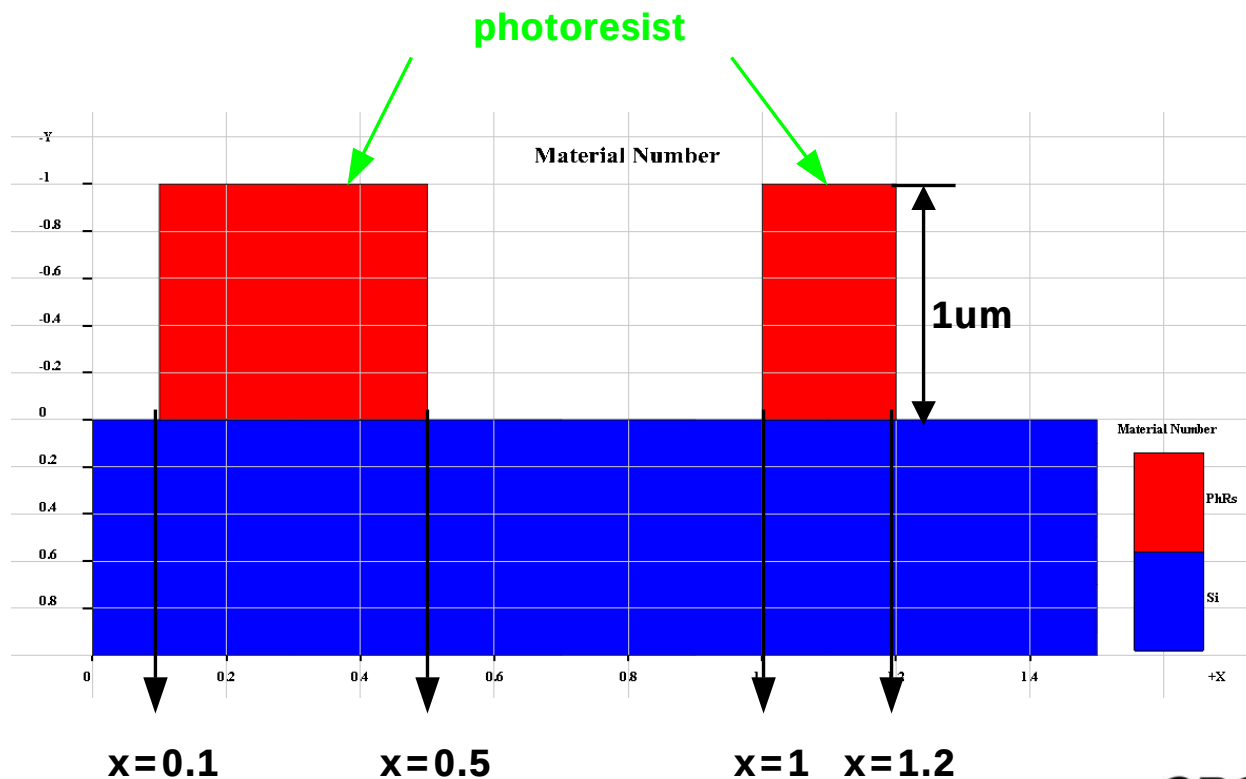
space between grid on outer edge

Mask

mask thick=1 x1.from=0.1 x1.to=0.5 x2.from=1 x2.to=1.2

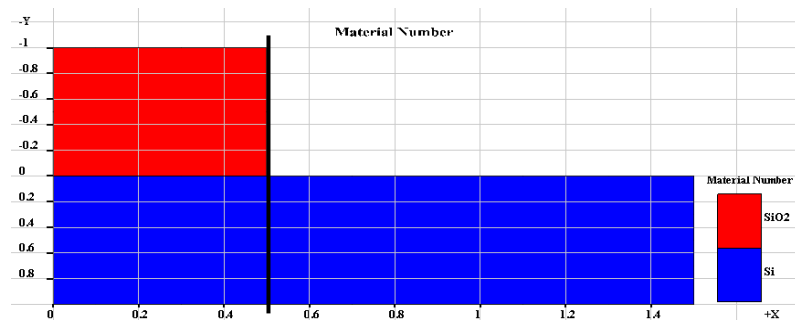


original

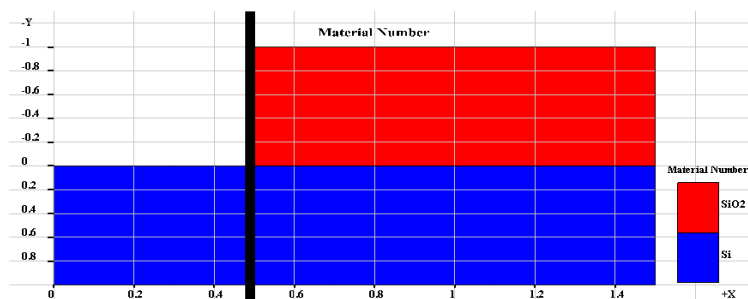


Etch: Left/right

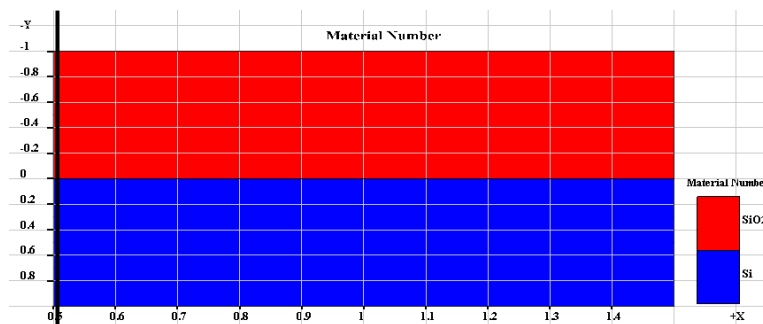
etch **oxide** right p1.x=0.5



etch **oxide** left p1.x=0.5

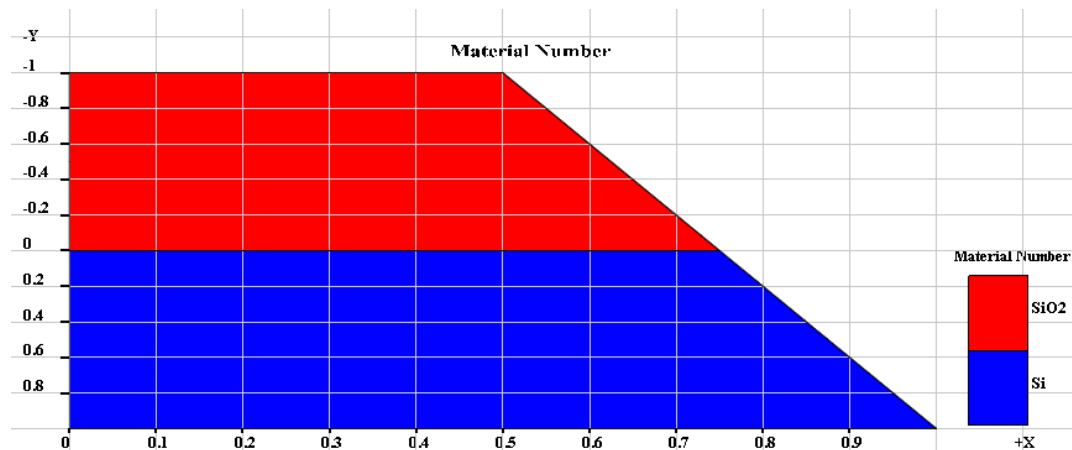


etch left p1.x=0.5

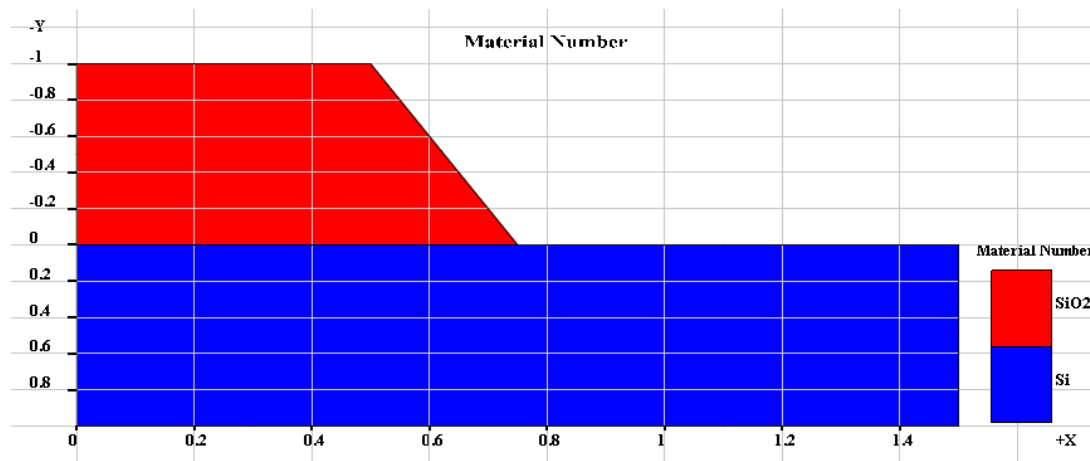


Etch: Left/right 2

etch right p2.x=0.5 p2.y=-1
p1.x=1.0 p1.y=1



etch right p2.x=0.5 p2.y=-1
p1.x=1.0 p1.y=1 **oxide**



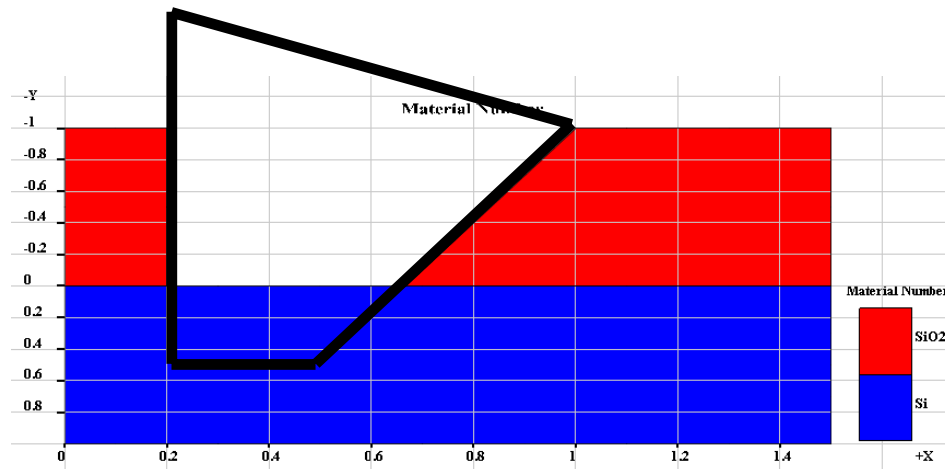
Etch: Start Cont Done

etch start x=0.2 y=-2 **oxide**

etch cont x=0.2 y=0.5

etch cont x=0.5 y=0.5

etch done x=1 y=-1

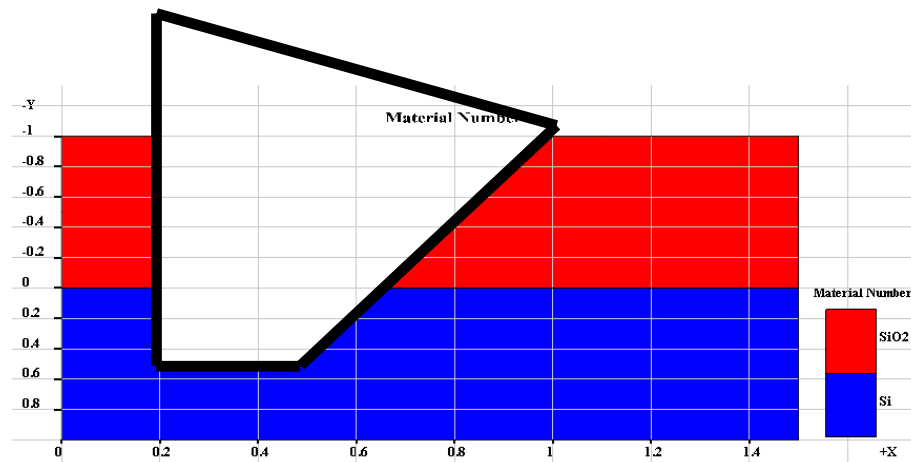


etch start x=0.2 y=-2

etch cont x=0.2 y=0.5

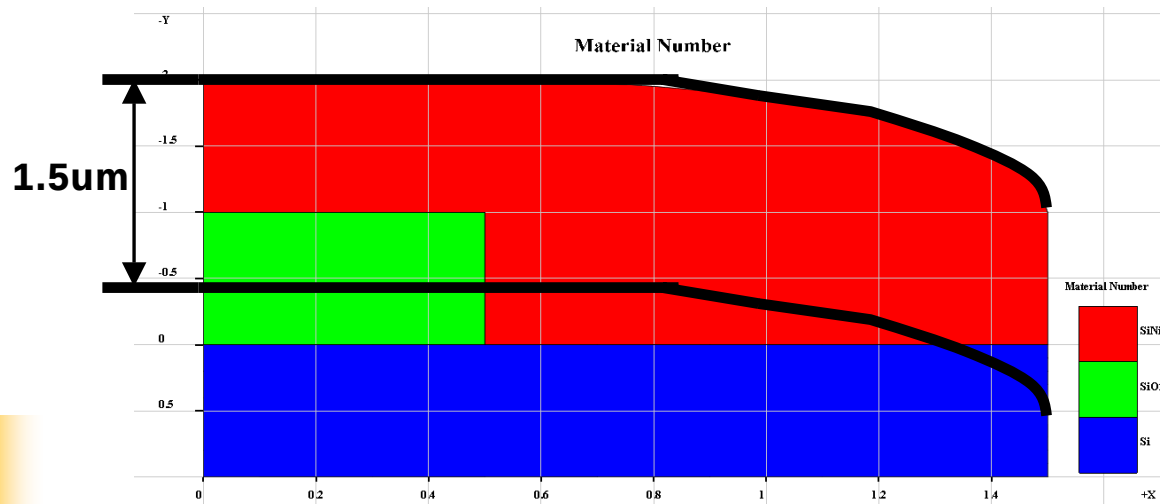
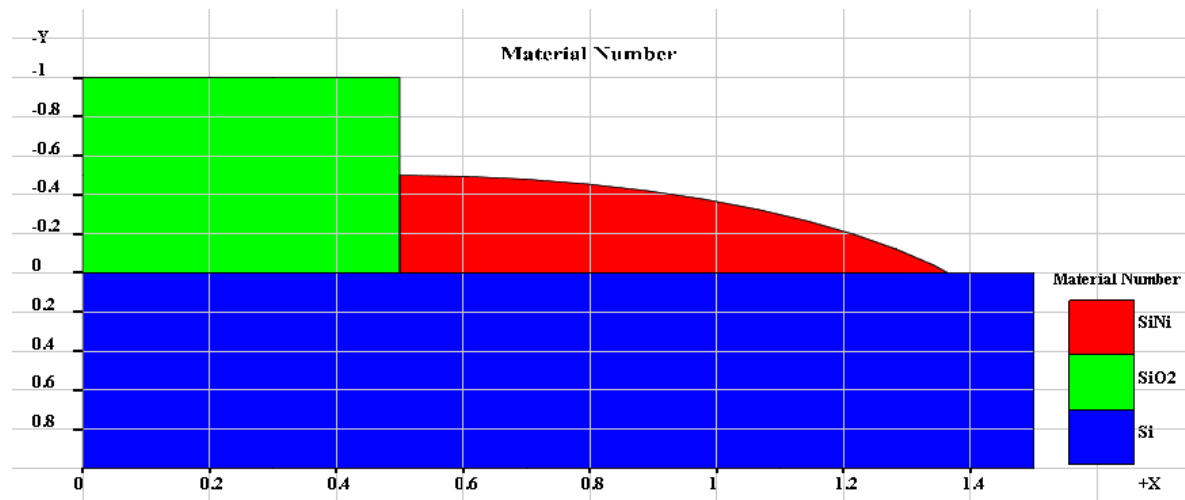
etch cont x=0.5 y=0.5

etch done x=1 y=-1



Etch: Dry

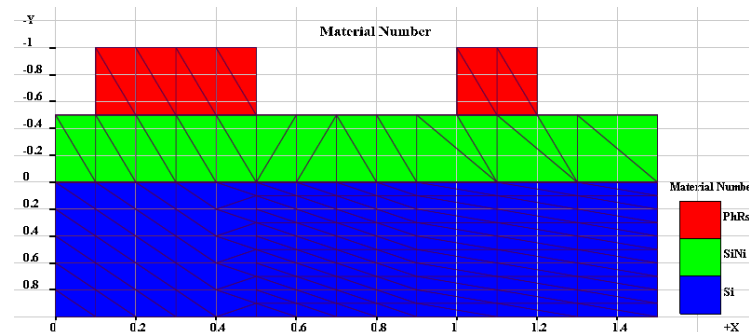
etch **dry** nitride thick=1.5



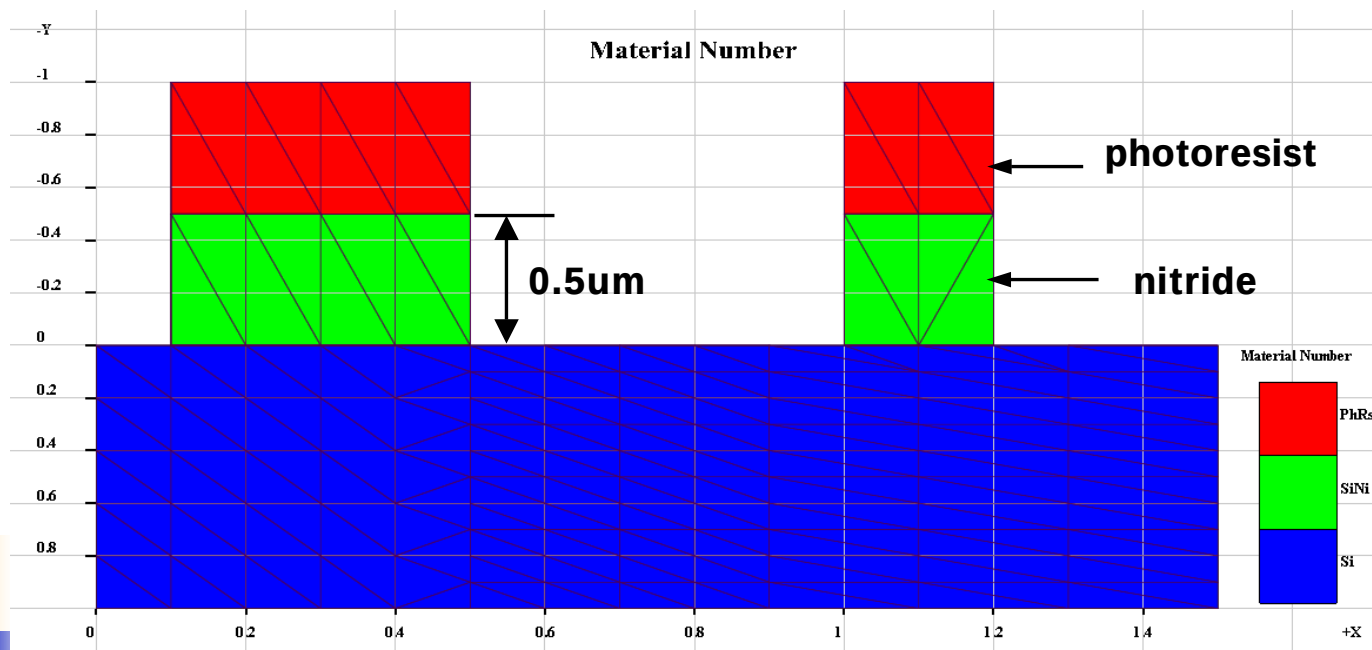
Etch: Avoidmask 1

etch avoidmask depth=0.9 nitride

Must follow mask



original



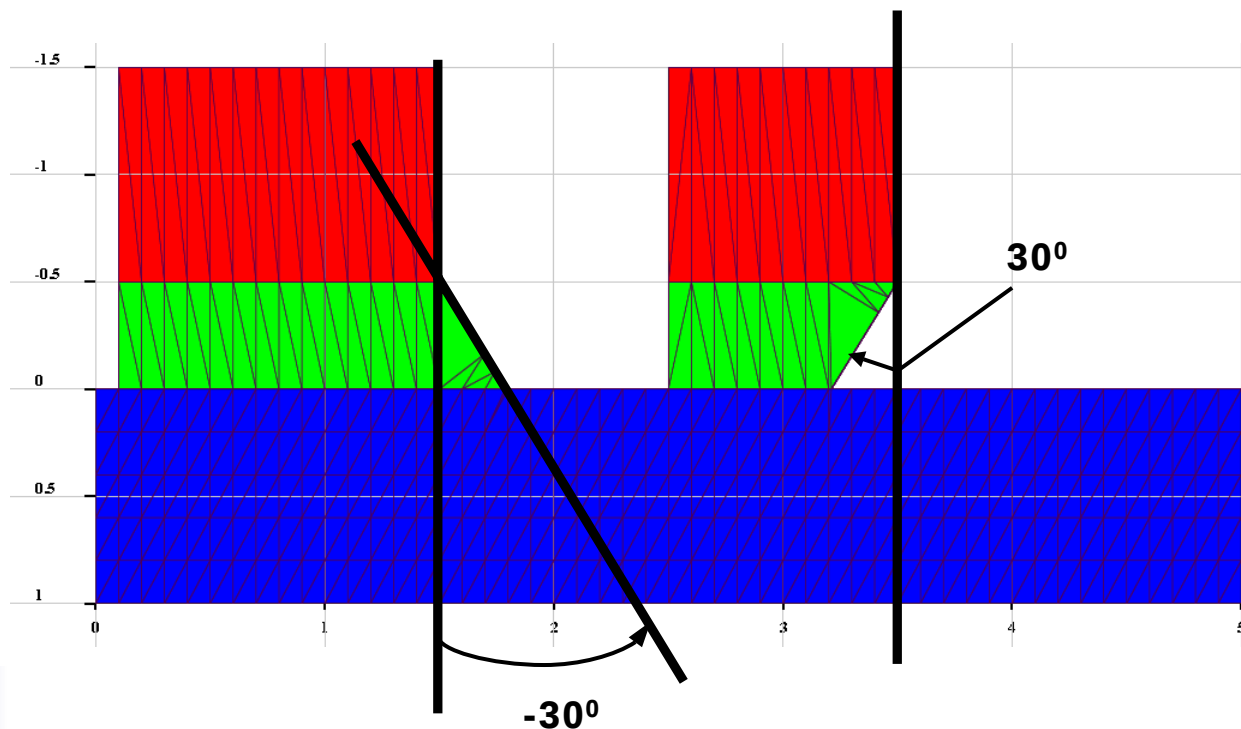
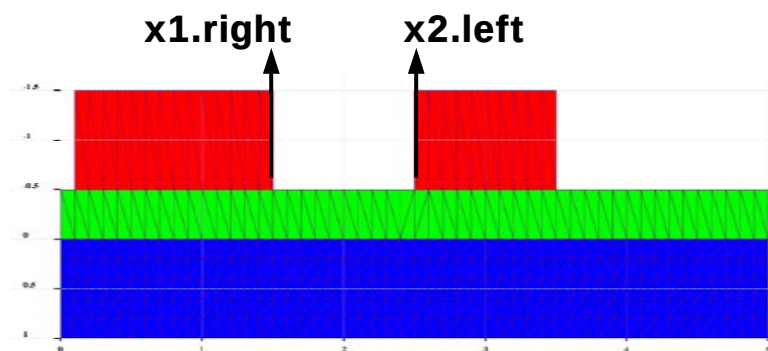
Etch: Avoidmask 2

mask thick=1

x1.from=0.1 x1.to=1.5 **x1.right.theta=-30**

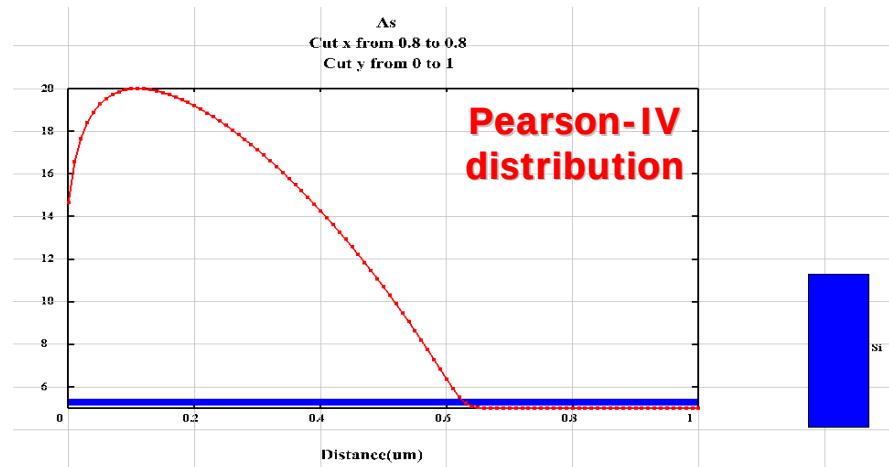
x2.from=2.5 x2.to=3.5 **x2.right.theta=30**

etch avoidmask depth=0.5 nitride

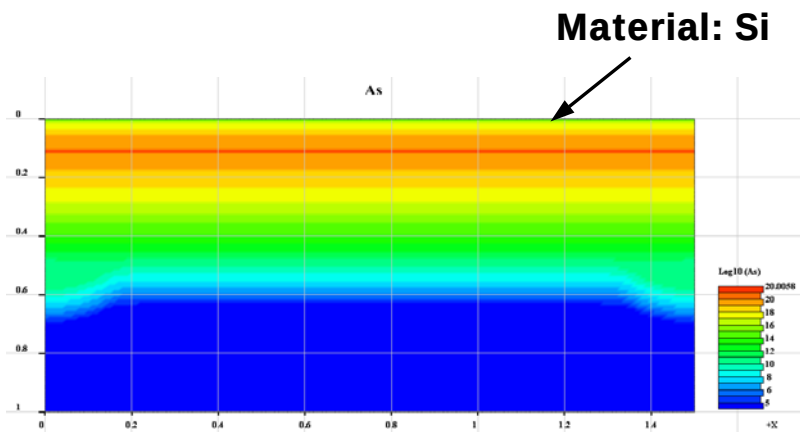


Implant

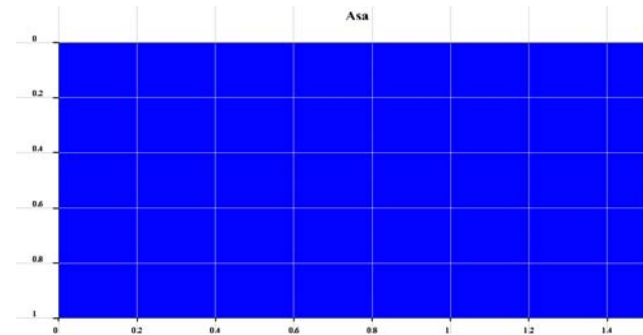
implant arsenic dose = 1×10^{15} ($1/\text{cm}^2$)
 energy = 200 [angle = 0] [rot = 0]
 (kev) (degree)



Arsenic at x = 0.8



Arsenic distribution



Active Arsenic distribution

Implant Table: sup4gs.imp

Imp Material Energy Parameters of different model

Dual pearson model

3	5	0.25	-0.0008	0.002	0.00001	4.00	0.002	0.005	0.005	6.3	688	0.25
---	---	------	---------	-------	---------	------	-------	-------	-------	-----	-----	------

Pearson model

3	5	30.	0.1011	0.0384	-0.480	3.2746	0.0569
---	---	-----	--------	--------	--------	--------	--------

Sims import model

3	57	1	sims	fsims_bf2_e1k_dose450e14.txt	-1
---	----	---	------	------------------------------	----

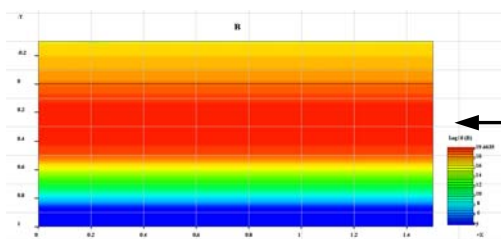
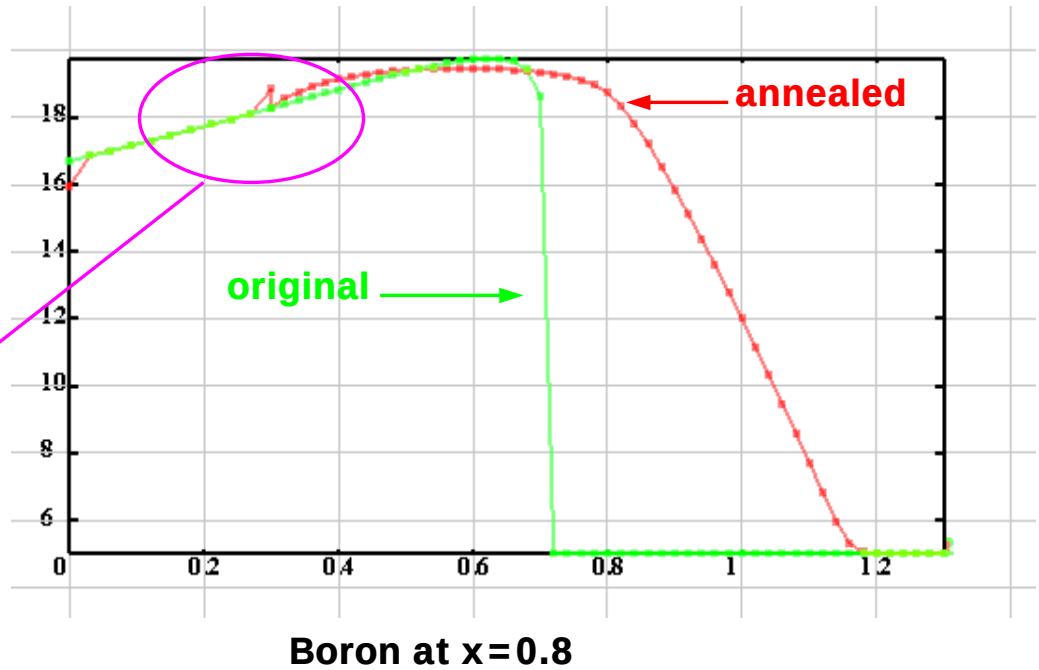
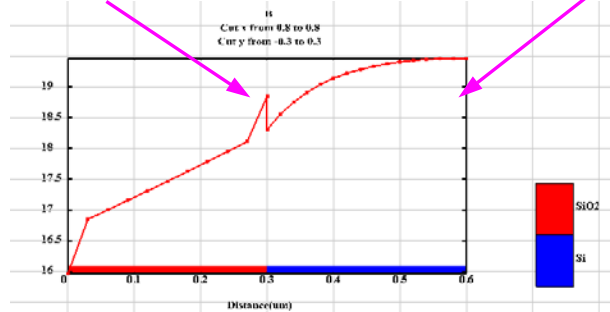
Gaussian model

2	31	10	0.0536	0.0293
---	----	----	--------	--------

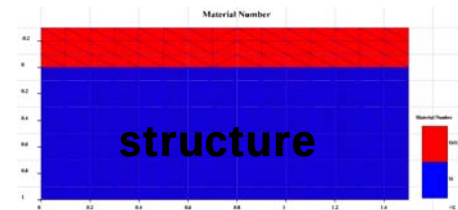
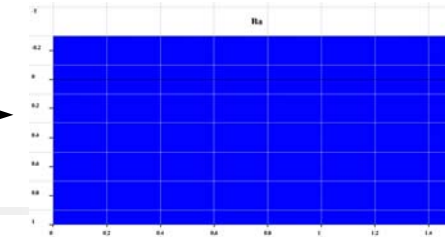
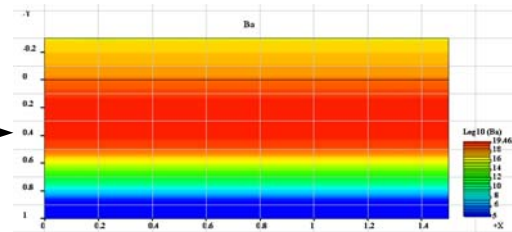
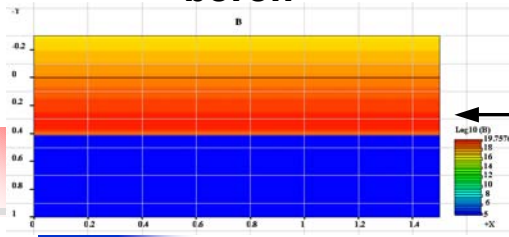
Diffuse

10min 1000°C
diffuse time=10 temp=1000
[final=1000] ↓
Temp can be ramped

Segregation at
 SiO_2/Si



boron

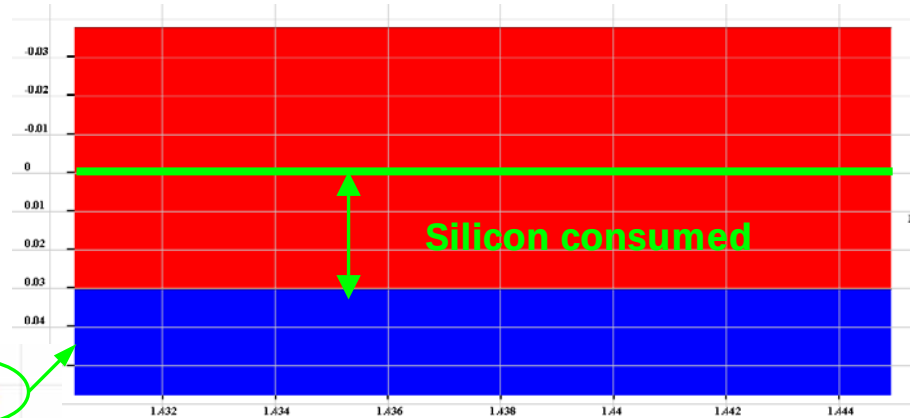
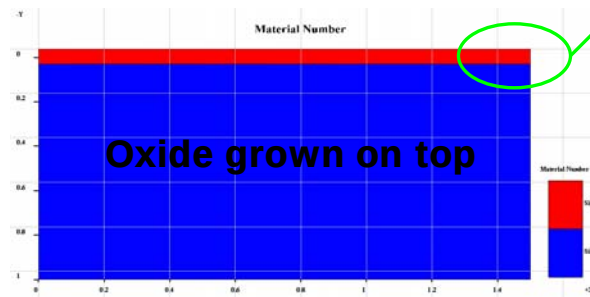


Diffuse With Oxygen: Dry Oxidation

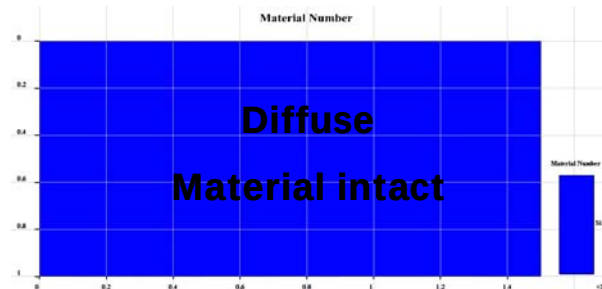
diff time=100 temp=1000

dry [pres=1]

partial pressure of O₂



diff time=100 temp=1000

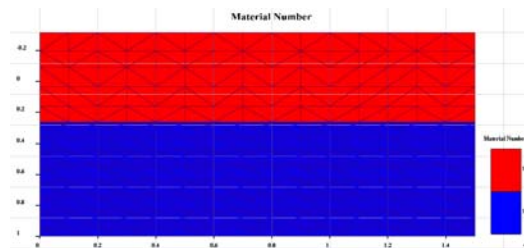


Diffuse With Steam: Wet Oxidation

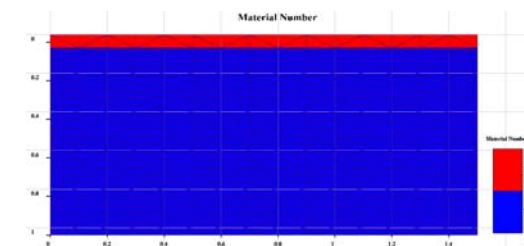
Partial pressure of steam

diff time=100 temp=1200 **wet** [pres=1]

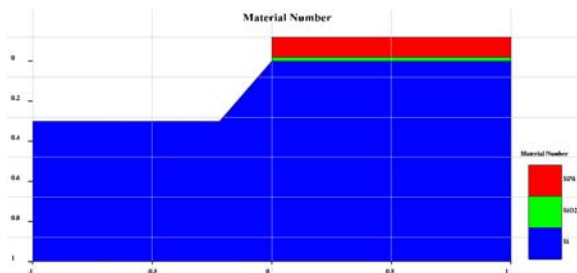
Thicker Oxide



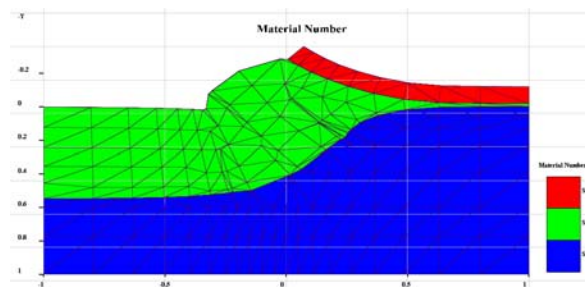
diff time=100 temp=1000 **dry** [pres=1]



Bird's beak by wet oxidation



original



meth compr

diffuse tim=90 temp=1000 weto2

Diffuse: Flow Control Model

diff time=100 temp=1000
 flow.control=t
 O2=1.5 [pres=1]

=

diff time=100 temp=1000
 dry pres=1

diff time=100 temp=1000
 flow.control=t
 steam=3 [pres=1]

=

diff time=100 temp=1000
 wet pres=1

diff time=100 temp=1000
 flow.control=t
 [pres=1] H2=4 O2=1

=

diff time=100 temp=1000
 wet pres=0.5

Gases: H2/O2/steam/HCl/other

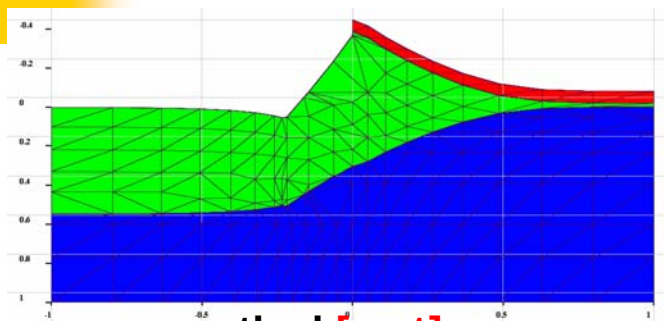
$2\text{H}_2 + \text{O}_2 \Rightarrow 2\text{steam}$

Wet and dry oxidation may occur simultaneously

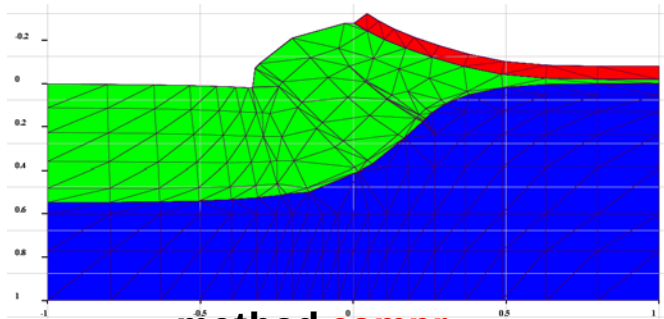
$P_{\text{steam}} = \text{steam} / \text{all gas} * \text{pressure}$

$P_{\text{oxygen}} = \text{oxygen} / \text{all gas} * \text{pressure}$

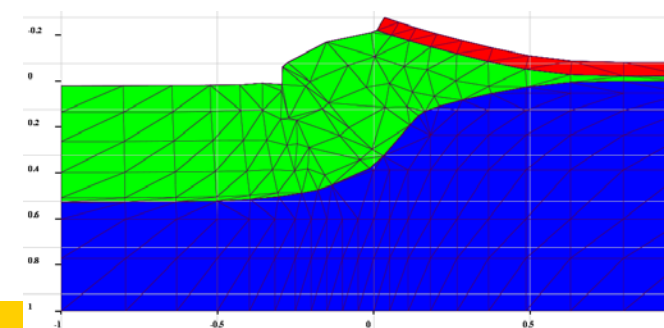
Method: Different Oxidation Model



method **[vert]**



method **compr**



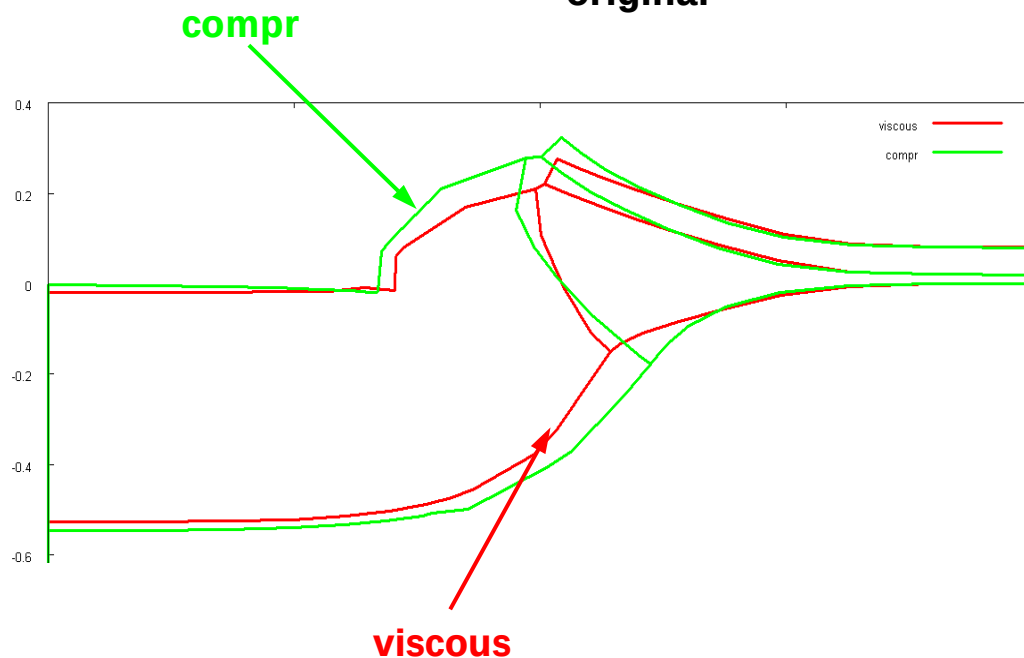
oxide stress.dep=t

method **viscous**

diffuse tim=90 tem=1000 weto2

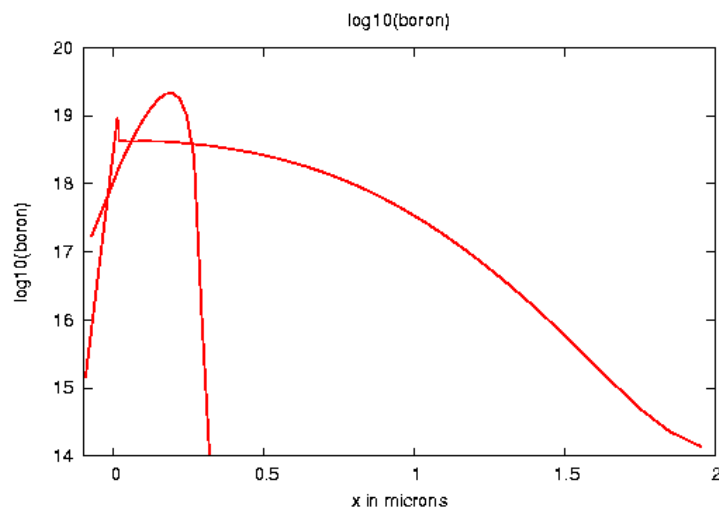


original

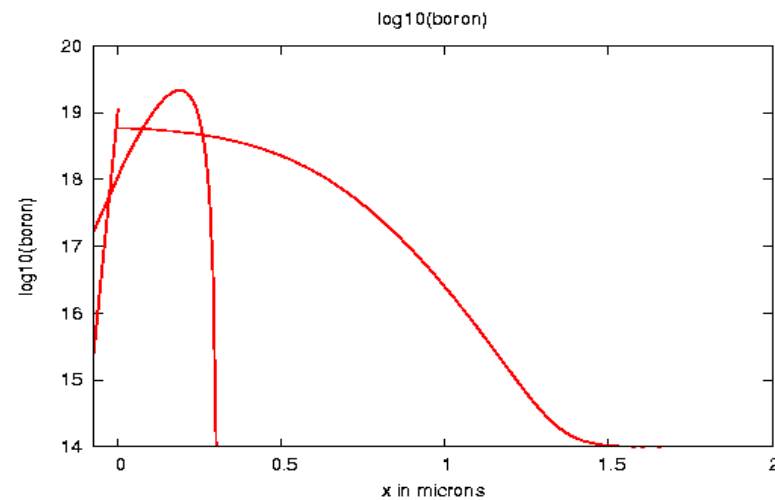


Method: Different Diffusion Model

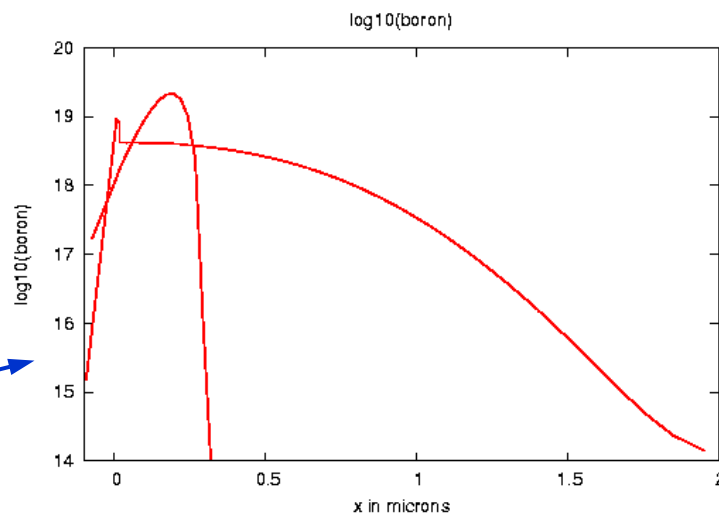
diffuse time=30 temp=1100 dry



method **two.dim**



method **[fermi]**



method **full.cpl**

oxygen enhanced
diffusion

Default Command Setting

File: suprem.key **-complete command listing**

card command

```
{  
keyword1=default;  
keyword2;  
}
```

File: modelrc **-default setting for physical parameters**

command keyword1=var1 keyword2=var2



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Input decks

- Output file numbering.
- Quit and restart.
- Mesh model setting using “option”.
- Regrid and mesh-repair.
- Going through an input deck for a simple nMOS.



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APSYS for MOSFET

- Temperature dependent impact ionization.
- Quantum mechanical models for carrier confinement and tunneling.
- Various vertical field dependent mobility models.
- Self-heating/thermal analysis with various types of thermal boundaries.
- Convenient transient/AC analysis suitable for various types of floating gate simulation.
- Parallel computing capability for unlimited mesh and multi-core CPU, suitable for Window64 and Linux84 OS.



Apsys input templates

- Getting started with Apsys: launching Apsys.
- Various modules and critical files.
- Going through an IdVd template for MOSFET.
- Using CrosslightView to get output data.